MFE 230GB · OCTOBER 8, 2026

Designing and Testing Global Trading Strategies

Two economically identified strategies — one in G10 rates and FX, one in emerging-market terms of trade — both using prediction markets as the alternative data the course asked for.

Short class presentation · Fall 2026 · Haas School of Business

One theme, two distinct books

Prices at horizons under a year are mostly demand and information flow, not scheduled macro. We trade that idea in two universes that cannot be copies of each other.

STRATEGY 1 · OECD

Policy Path

G10 FX forwards plus G7 duration. Prediction-market changes in the Fed path, recession odds, and elections are treated as demand shifters — the Topic 4 / Topic 5 object — not as a second macro forecast.

Universe AUD, CAD, CHF, EUR, GBP, JPY, NOK, NZD, SEK; US/DE/JP 10-year

Horizon Weekly rebalance, 10% vol target

Alt data Kalshi + Polymarket ΔP and cross-platform disagreement

Condition G10 carry on global risk. Do not run it raw — it has not paid since 2010.

STRATEGY 2 · EM / NDF

Terms of Trade

Long oil-exporter EM FX versus oil-importer EM FX, but only when the terms-of-trade shock is on — Brent moves, or prediction-market odds of a supply disruption jump. Same mechanism as Topic 3; different assets, NDFs, and crash risks.

Universe BRL, CLP, COP, ZAR vs INR, THB, CNY via NDF/spot

Horizon Weekly, dollar-neutral, 12% vol target

Alt data Geopolitical / OPEC contracts + Brent; not just the oil print

A static exporter–importer book does not pay. Timing the ToT shock does.

What we take from each topic

01	**Instruments**	Spot, forwards, NDFs, and the spread/liquidity gap between G10 and EM. Strategy 2 is an NDF book on purpose.
02	**Regimes**	EM is not a dirty float of G10. Reserves, the trilemma, and convertibility change both the signal and the crash.
03	**Parities**	CIP holds; UIP and short-run PPP do not. Carry is a downside-risk premium. Fiscal news can flip the yield–FX sign. A 10% ToT shock appreciates the real exchange rate.
04	**Order flow**	Macro explains ≤5% of monthly FX variance; one order-flow series explained 64% of daily DM/USD. News is priced in the trade, not the release window.
05	**Demand**	Latent demand accounts for 82% of exchange-rate variation. Elasticities are 1–30, not thousands. The dollar's specialness is ~1.4% a year.

How we read the assignment

01 Mechanism first

A signal enters the book only if we can say who is trading and why the price must move. Backtests come after that sentence, not before.

02 Two real universes

G10 can use spot, forwards, and cash bonds. EM uses NDFs and lives with convertibility and 10–20 bp costs. They are not the same strategy with a different ticker.

03 Alternative data that is a shifter

Prediction-market ΔP is the demand/information shock. Disagreement across Kalshi and Polymarket is a risk-off switch, not another forecast.

04 Risk that matches the story

Strategy 1 dies in a carry crash and a fiscal-dominance week. Strategy 2 dies in an oil collapse and an EM liquidity squeeze. We measure those, not just max drawdown.

STRATEGY 1 · DEVELOPED MARKETS

Policy Path

The hypothesis is about who is trading

≤ 5%

of monthly FX variance explained by measured macro fundamentals

64%

of daily DM/USD changes explained by interdealer order flow in Evans–Lyons

82%

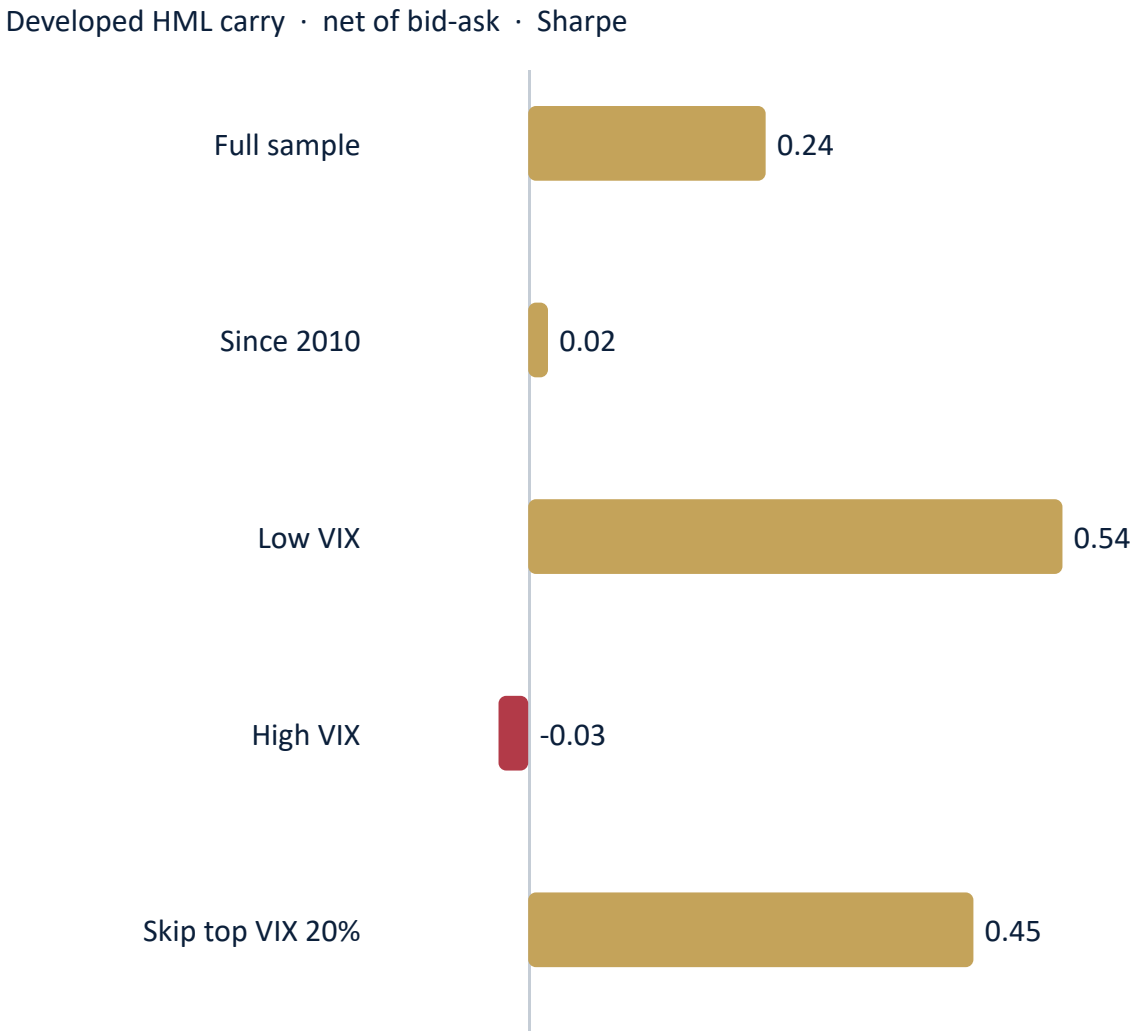
of exchange-rate variation attributed to latent demand in Koijen–Yogo

What we claim

A rise in prediction-market odds of a Fed cut, a recession, or a fiscal/election shock is a public but not fully priced demand shifter. It should move G10 FX and duration the way order flow does: immediately, and outside the 20-minute announcement window.

Why it is not a second Taylor rule: scheduled payrolls surprises move USD/DM ~13 bp against a 66 bp daily sigma (Topic 4). The information that matters arrives through trading. Prediction markets aggregate that information before the FOMC statement, and inelastic dollar demand (Topic 5) maps the flow into a persistent price.

Unconditional G10 carry is the wrong trade



Lustig–Roussanov–Verdelhan HML, developed, net of spreads. Same series as Problem Set 1.

Full sample Sharpe 0.24, max drawdown −39%.

Since 2010 the mean is ~0. The risk premium did not vanish — the crashes and the post-GFC compression ate it.

Low-VIX months: Sharpe 0.54. High-VIX months: −0.03. Oct 2008: −13.8% in one month.

This is Topic 3's downside-risk CAPM, not a reason to abandon carry. It is a reason to turn it off when global risk and policy-path odds say a crash is being priced.

Three signals, one book

A Carry-to-vol

3-month rate gap over 3-month realized FX vol, G10 vs USD. Ranked cross-section, long top three, short bottom three. Topic 3 carry-to-vol, not raw carry.

B Policy-path ΔP

Five-day change in Polymarket and Kalshi probabilities of a next-meeting cut, a 12-month recession, and the relevant election/fiscal contract. Trade the change, never the level.

C Disagreement

|P_Kalshi − P_Polymarket| and the contract's implied vol. When disagreement is in the top quartile, cut gross exposure by half. This is the brief's 'disagreement across platforms' clause.

From signal to position, every Friday

1

Rank G10

Sort by carry-to-vol. Long 3 / short 3, inverse-vol weights, dollar-neutral on the six legs.

2

Read ΔP

If 5-day ΔP(cut) > +5 pp: add long US duration and short USD vs EUR and JPY. Opposite if ΔP(hike) > +5 pp.

3

Safe-haven overlay

If ΔP(recession) > +5 pp: flatten carry, long JPY and CHF, short AUD and NZD. This is the Topic 3 crash-hedge.

4

Fiscal sign check

If local 10y yield and the currency move in opposite directions that week, treat it as fiscal dominance (Truss / Japan 2026). Short the currency and the long bond together.

5

Scale

Gross FX 2×, duration 1×, then vol-target the book to 10% annualized. Zero out if VIX is above its 80th percentile or disagreement is elevated.

Mechanism test, not the final backtest

0.45

Sharpe of developed HML if we simply skip the top VIX quintile, versus 0.21 unfiltered (1990–2021, net)

–13.8%

Worst month: October 2008. The other crash cluster is 1998 / 2007–09 — carry's priced risk, not noise

1.4%

Dollar specialness in Koijen–Yogo: foreign latent demand for US assets is worth ~1.4% a year. A policy-path shock hits an inelastic dollar book

When Policy Path loses money

Carry crash

VIX spike, AUD/NZD/NOK collapse, JPY squeeze. We measure HML return on days when the global dollar factor is in its 5th percentile. The filter is supposed to cut this; if it does not, the strategy is rejected.

Fiscal dominance

Yields up, currency down (UK 2022, Japan 2025–26). A policy-path long-duration overlay is the wrong side. The sign check in step 4 is the hedge; we still report the unhedged loss.

Announcement catch-up

If almost all of the PM signal's predictive power sits inside the FOMC window, we are not discovering order flow — we are trading the statement. That falsifies Topic 4 as the mechanism.

Crowded PM

Prediction-market depth is thin versus FX. If the ΔP has already been printed in 2-year yields that morning, we are late. We require the PM move to lead, not lag, the 2s.

STRATEGY 2 · EMERGING MARKETS

Terms of Trade

A terms-of-trade shock is a demand shock for the currency

Topic 3: a 10% terms-of-trade improvement appreciates the real exchange rate and raises output. Oil exporters (NOK, BRL, MXN, CAD) gain when Brent rises; importers (KRW, JPY, TWD, THB, INR, EUR) lose — after stripping out the dollar's own swing.

Why it should predict

The oil move is public. The FX move is not instantaneous because EM demand is inelastic, NDFs segment the market, and the investor base (Topic 5) adjusts with a lag. Order flow in the NDF is how the ToT news gets into the price (Topic 4).

Why not always-on

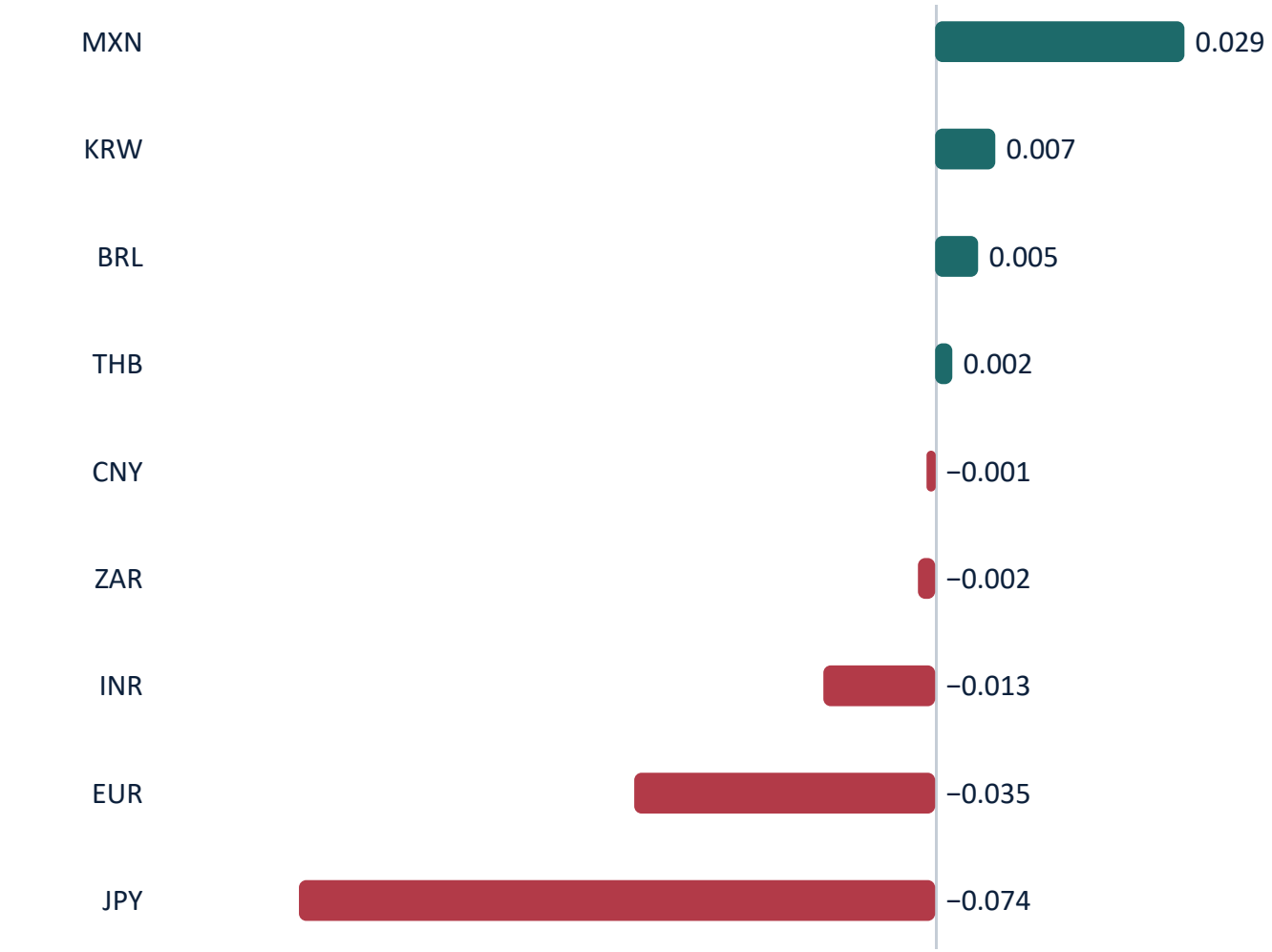
A static long-exporters / short-importers book is just another high-beta EM carry. In our 2010–2026 weekly sample that book's Sharpe is ~0.04. The premium is for bearing the shock when it happens, not for sitting in BRL.

Role of prediction markets

Brent is the realization. Polymarket/Kalshi contracts on Iran, OPEC cuts, and shipping disruption are the probability of a ToT shock before the print. ΔP should lead both oil and the exporter FX.

Dollar-controlled oil betas line up with the lecture

Weekly dFX = a + b dBrent + c dDollar · 2010-2019



FRED H.10 and Brent, weekly, 2010–2019. Same specification as the Topic 3 oil-shock slide: strip out the broad dollar so we are not just trading DTWEXBGS.

MXN and BRL load positively. JPY, EUR, and INR load negatively. ZAR is a commodity currency that still loads on the dollar more than on oil — which is why we estimate betas rather than using a textbook exporter list.

Betas are estimated in sample and frozen for the 2020–2026 out-of-sample book.

The EM book, and a first out-of-sample look

Trading rule

Each Friday, re-estimate 3-year rolling oil betas, dollar-controlled.

Long the two highest-beta EM names, short the two lowest, equal dollar, NDF where needed.

Hold the LS only if last week's Brent return is positive or 5-day ΔP(supply disruption) > +5 pp. Flip the sign if oil is down and ΔP is not rising.

Vol-target 12%. No pair > 35% of gross. Skip a name if 1-month realized vol is above its 95th percentile (convertibility / crash filter).

Preliminary · oil-timed LS only

This uses Brent timing, not yet the prediction-market overlay. It is the mechanism check.

0.28

Sharpe, 2010–2026, oil-timed

0.17

Sharpe, 2020–2026 out of sample

–13%

Max drawdown, OOS

~0.04

Sharpe if we never time — rejected

When Terms of Trade loses money

Oil crash, not a ToT boom

March–April 2020: Brent −35% then −23% in consecutive weeks. BRL and ZAR both sold off with global risk. A long-exporter book is a high-beta book on those weeks. Timing helps; it does not eliminate the left tail.

Country risk dominates oil

A Brazil fiscal scare or a South Africa load-shedding / politics week can move BRL/ZAR more than Brent. We report a country-dummy regression and cap any one name at 35%.

NDF liquidity and convertibility

INR, THB, CNY, CLP trade as NDFs (Topic 1). Spreads gap when the dollar squeezes. We haircut returns 10–20 bp round-trip and show a version that drops any pair whose 1-month vol has doubled.

Dollar wave

If the LS is secretly a short-dollar trade, we have not isolated ToT. Every result is reported dollar-neutral and after hedging DTWEXBGS. Residual dollar beta must be near zero.

Same theme, different crashes

	Policy Path · OECD	Terms of Trade · EM
Mechanism	Inelastic dollar demand + order flow around policy-path information	ToT shock into segmented NDF demand
Alt data	Kalshi / Polymarket Fed, recession, election; disagreement	Geopolitical and OPEC contracts + Brent
Instruments	G10 forwards, US/DE/JP 10-year	EM NDFs (BRL, INR, THB, CNY, CLP)
Horizon	Weekly, 10% vol target	Weekly, 12% vol target
Turns off when	High VIX, high PM disagreement, fiscal-sign flip	Oil crash, country-risk spike, NDF vol doubling
Falsifier	PM ΔP only predicts inside the FOMC window	LS is just a dollar trade, or always-on equals timed

What we will report, including the tests that can kill it

Performance

Cumulative excess return, mean, vol, Sharpe, max DD, turnover, hit rate — gross and after G10 2 bp / EM 15 bp round-trip.

Out of sample

Signals and betas frozen through 2019; 2020–2026 is walk-forward. A second split: train to 2016, test 2017–19, then 2020–26.

Placebo

Shuffle PM ΔP dates; the Sharpe must collapse. Always-on carry and always-on exporter LS are the published nulls.

Lead/lag

PM ΔP must lead 2-year yields (S1) and Brent (S2) at least some of the time. If it only lags, we are not using alternative data.

Risk overlays

Carry-crash dummy, oil-shock dummy, global dollar dummy, country dummies. Report beta, not a story.

Concentration

Share of P&L from top pair, top month, and top contract. A strategy that is one week in 2026 is not a strategy.

Data, costs, and the other deliverables

Prices	FRED H.10 FX, Brent, VIX, OECD 3-month and 10-year yields. Lustig currency portfolios for the G10 carry history. Bloomberg/Refinitiv NDFs for the live EM book.
Prediction markets	Kalshi and Polymarket historical contracts: Fed decision, US recession, elections, Iran/OPEC/shipping. Store raw JSON. Document every AI prompt in the README.
HTML + package	Self-contained page: idea, signals, equity curves, crash tables, lead/lag. Replication: data or download scripts, all notebooks, costs file, and a one-page README.

What we want you to remember

Carry and terms of trade are not dead. Unconditional versions of them are. We trade the shock that the course actually identified — a demand and information shock — with prediction markets as the alternative data, in two universes that fail in different weeks.